

Social Control Interaction Framework: Design to Technically Support a Group of Users in Making Control Decisions Together

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Figure 1: Visual demonstration of the social situation in the car (left) and in the living room (right) where users want to make decisions together by interacting simultaneously with entertainment services.

ABSTRACT

Entertainment is enjoyable and helps people to feel socially connected with others. However, the control of entertainment applications in group situations is often restricted to one person. This especially limits group decision-making where people sit in close proximity such as in the living room or inside a car. Restricting users in group decision-making can cause interpersonal conflicts and frustration. To overcome these aspects, my PhD research aims to articulate design principles to support social control. I specifically focus on distributing control across group members to support collaborative group decision-making and enhance social connectedness. To achieve this, I will design entertainment-oriented social control services for the car and the living room. Based on the exploration and designs for the car domain, I will transfer the gained knowledge to the living room, to verify and generalize my findings. This should support in drawing an overall conclusion for the design of general social control entertainment services. Thus, my dissertation contributes to the understanding of the design for future, entertainment-oriented, distributed control systems to best support group collaboration.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative interaction**; **Collaborative content creation**; *Interactive systems and tools*.

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KEYWORDS

Social control, collaboration, distributed control, decision making, interactive system, social connectedness, entertainment services, automotive, smart home

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1 CONTEXT

I undertake my PhD studies at the Department of Industrial Design at the Eindhoven University of Technology (the Netherlands), under the supervision of Prof. Dr. Regina Bernhaupt, Prof. Dr. Bastian Pfleging, and Dr. Bahareh Barati. The research topic I am investigating is industry-based and funded by the company ruwido austria GmbH. My PhD track is a four-year program (full time), which I started in October 2019 and which I am on track to complete in October 2023.

2 INTRODUCTION

The way we consume media has changed considerably in the last 20 years. With the development of video and audiotapes, media have established themselves as consumer goods. Through technical innovations and digitalization, media content is nowadays accessible at any time on nearly every digital device, independent of the user's location (e.g., TV, smartphone, laptop). Established innovations, especially in the living room or inside the car are digital movie libraries or (shared) music playlists (e.g., [15, 16, 37]). According to statistics, the majority of media content gets consumed within a social environment, since the average European household comprises

2.3 people [11], and even in every second car ride, the driver is accompanied by at least one passenger [28]. However, most digital systems for entertainment leave the selection of media content still to a single person, as interaction concepts only allow for input from one user at a specific time. This especially applies to using a remote control to operate the TV or having a single touch-screen mounted inside the car. Consequently, people often sit nearby at home or in the car not being able to participate in the control of entertainment services.

Previous research reports that limiting people in the participation of decision-making processes causes social conflicts, frustration and can also lead to vandalism [19]. A way to overcome these issues is to technically support users in making decisions together [23, 30]. Concerning entertainment services, Kenton et al. show that democratic music selection in public spaces enhances social interaction and communication [29]. Also, watching movies synchronously with friends establishes a social bond [16]. In addition, Ammarapala et al. report that collaborative decision-making does not only get influenced by people's personalities [2]. Instead, social connectedness has a major influence on achieving a goal together [2]. Whilst some research has focused on connecting users virtually (e.g., [9, 31]) or in public spaces (e.g., [15, 29]), little is known how to technically design collaborative, entertainment systems to best support group control decision-making in a closed environment such as in the car or the living room.

With my thesis, I want to fill this gap by investigating how to design interactive, entertainment services to enable a group of people to control together and to bring them socially closer. To understand, design, and evaluate these objectives of social control, I will focus on entertainment in the automotive and (smart) home domain, since entertainment is well established there. More precisely, I am going to focus on people sitting inside a manually driven car or in the living room. Thus, with my work, I contribute to the understanding of how to design interactive control systems to best support collaborative group decision-making and social connectedness. In the following, I provide an overview of key publications which frame my overall research objectives. Besides that, I outline my current contributions, the planned next steps, and how my overall PhD program contributes towards HCI research.

3 BACKGROUND & KEY PUBLICATIONS

Social interaction occurs everywhere, especially in situations where people collaborate to achieve a common goal [35]. The effectiveness of a collaborative activity gets influenced by people's personalities [36], their values [3], the overall perceived social connectedness [22], the size of the group [33], and decision-makers involved [2]. Due to the complex, dynamic structure of collaboration in a team setting, the ability to achieve a common goal depends strongly on the role differentiation (e.g., contribution possibilities or authority levels) [27] and the presence of decision-makers [24]. Social psychology defines four major types of decision-makers within the decision-making process [2, 24]. First, the *multiple decision makers*, where group members do not have equal authority in making a decision, but none of them has enough authority to make all decisions alone [24]. Secondly, *group decision makers*, defined as each group member having equal weight in making decisions [24].

Thirdly, *team decision makers* where an individual decision-maker has the authority to make final decisions under negotiated outcome [24]. And lastly, the *individual decision makers*, where group members decide alone with the focus on achieving a goal without group negotiation [2].

To design interactive systems that enable authority levels and support decision-making, the focus lies on users' abilities and perceived responsibility in interplay with the systems' provided authority and control [12]. This human-machine cooperation model has been applied in different dynamic, safety-critical application fields such as air traffic control (e.g., [26]), air vehicles (e.g., [10]) or driving assistance (e.g., [1, 8, 14]). Overall research shows that to best support collaboration, an interactive system needs to be centered around the group of users [32] and enable any group member to participate actively [20, 24]. To be able to design for individual control contribution and social connectedness, a clear balance of the presence of decision-makers [24], a reasonable group size [2, 34], and individual access to functions is key [12]. However, current literature provides little about how decision-making processes [24] and role differentiation (=shared control) [12, 27] can be combined to design interactive, collaborative systems to support group collaboration and enhance social connectedness. With my PhD, I intend to fill this gap and investigate how we can technically design for social control by distributing control across group members. Therefore, I focus on entertainment services which are used in the automotive and (smart) home domain.

4 RESEARCH OBJECTIVES & METHODOLOGY

To best design for social control, the dissertation is guided by user-based research. I am following the user-centered design approach [13] since it enables me to identify user-specific issues, motivations and helps me to design systems that fit users' needs. The cyclic nature, which includes feedback and evaluation of design ideas, enables continuous improvements and allows me to verify my concepts in terms of collaborative support.

My main research question (RQ), based on the research gaps outlined above, focuses on *how to design an interactive, entertainment service to distribute control with the goal to support collaborative decision-making*. To answer this main RQ, my research focuses on answering the following sub-RQs (also illustrated in Figure 2).

RQ1: What makes a collaborative, social experience in the living room/in a manually driven car? To understand the users' objectives in these two domains (automotive, home), I conducted literature reviews with a focus on users' needs, motivations, and (social) experiences. To verify and extend the gained knowledge I am to conduct semi-structured interviews and focus groups.

RQ2: How can we enable social control for entertainment services? For the technical design of social control, I aim to consider the four different ways of group decision-making [2, 24] in combination with the different user authority levels [12]. To assess the impact on collaborative group decision-making and social connectedness, I will focus on entertainment use-cases in the automotive and (smart) home domain. The specific use-cases are defined based on the results from RQ1. Since entertainment is well established in these two domains, it allows me to investigate how social

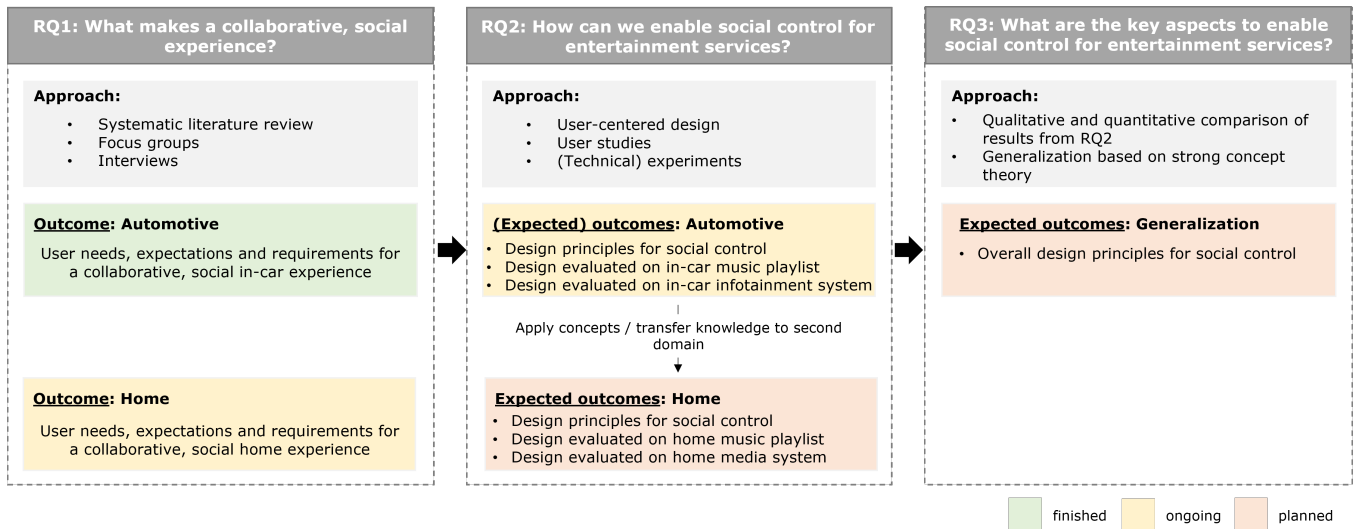


Figure 2: Research questions and methodology of my PhD studies

control can be enabled for specific actions of a single entertainment application (use case 1: music playlist, e.g., play, pause functions) as well as for different menus of a whole entertainment system (use case 2: infotainment system, e.g., access to radio or music library menu). First, I start with applying the social control concepts to the two use cases in the automotive domain. After improvement of the concepts based on insights derived, I will transfer the concepts and knowledge to the living room to establish my findings.

RQ3: What are the key aspects to enable social control for entertainment services? I will systematically compare the results from the four experiments (two automotive domains, two smart home domains). An expected outcome will be a framework for the design of social control applications. It will outline recommendations and guidelines to technically enable social control and how to support collaborative group decision making as well as social connectedness for interactive (entertainment) systems.

5 RESULTS & CONTRIBUTION TO DATE

In the following, I present the research that I have conducted over the past 20 months. I outline the main results of the literature review and interviews that I conducted with regard to the in-car experience. In addition, I report on case studies that I did to support social control in the car. Moreover, I detail how my work addresses gaps in previous literature.

RQ1 – Literature Review & Understanding the Social Experience in the Car: In the first phase of my doctorate, I conducted a literature review and a repertory grid interview study to understand the passenger experience when riding in a manually driven car. A scoping review was carried out to generate insights into passengers' needs and motivations to understand how to support passenger experience. In three databases (*ACM Digital Library*, *Scencedirect*, *Springer*), I identified 68 papers that describe passenger-based applications in the manually driven car domain, while only 8 papers reported information connected to passengers' experiences. I used the reported aspects of these papers and categorized them

into motivational factors [38], basic human needs [25], or additional properties. The resulting review outlines the importance of belongingness (e.g., conversations [21, 39], shared accomplishments [21], co-habitation [21], trust [18], perceived hospitality [21]), intrinsic motivation [38] (e.g., using gadgets to avoid boredom, entertainment, observe the surroundings) and extrinsic motivation [38] (e.g., arrive safe, overall safety). Since the information from the literature review is rather scarce, I conducted a repertory grid interview study [7] to explore the passenger experience in more breadth and depth. Overall, I identified nine main categories that constitute a convenient passenger experience [7]. Based on these categories, I outlined eight design recommendations to support passenger experience. A major factor lies in the social situation when riding a car. Passengers especially want to feel socially connected to others, develop the feeling of belongingness, strive for active participation (e.g., assisting the driver, interacting with a device or system), and want to get entertained (e.g., music, exploring the landscape) when sitting in a car. In the context of my PhD program, the results of the repertory grid interview study affected the general design of social control (see Figure 2). Based on the gained insights, I also specified the use cases for the evaluation of social control in the car.

RQ2 – Case Studies Automotive Domain: In the second phase of my PhD, I conducted case studies with a focus on in-car interactive concepts for passengers. The purpose of the case studies was to build on my previous work [7], get familiar with designing for the automotive domain and to identify concrete applications to support social control in the car. First, I combined the need for entertainment and exploration of the outside world with the possibility to provide control over certain in-car functions. Thus, I developed an augmented reality-enabled interactive car door [4] which allows passengers to interact and engage with passing attractions. With a high focus on the exploration of outside attractions, I designed an extension of an in-vehicle infotainment system (IVIS) that encourages drivers and passengers to take breaks by recommending stops that fit occupants' interests [5]. Since the

social experience plays a major role in my studies, the last case study focused on driver-passenger collaboration. Therefore, I developed a physical, interaction concept that allows drivers to request assistance by delegating IVIS functions towards the passenger [6]. The case studies showed me possibilities of interaction inside a car and outlined that the novelty of a concept majorly influences the (social) experience of users. Thus, developing and evaluating social control concepts requires a familiar use case to avoid bias due to the novelty of innovations.

RQ2 – Use Case 1 Social Control in the Car: Based on the knowledge gained through the case studies, in combination with insights from literature about group decision-making [2, 24] and user authority levels [12], I designed five social control concepts. These five concepts distribute control across group members to support group work based on joint, hierarchical, individual (two different concepts), and single control decision-making. I first evaluated these concepts inside the automotive domain by applying them to use case 1, a shared music playlist. Since this is a well-established entertainment use case, it enables to test the social control concepts without novelty bias. For the empirical assessment, I conducted a mixed-subject experiment ($N = 27$) inside a parked car. Results show that joint control decision-making and single control decision-making connect people socially best. They encourage team communication and support negotiation. Individual decision-making instead limits reaching the group goal, since communication is low and group members can overrule others' decisions (details missing due to paper under review). The outcomes of this experiment are of high importance for the next phase of my PhD research. The investigated concepts are the foundation of my social control framework that I want to develop and thus the results are highly relevant to further develop social control for the second in-car use case and the smart home domain.

6 CURRENT AND NEXT STEPS

Having presented the results and current research activities of my doctorate so far, in the following section I outline ongoing projects and the next steps planned. The planned studies present my latest research plan, leaning onto the described approach above.

RQ2 – Use Case 2 Social Control in the Car: To strengthen the social control concepts, I am currently preparing my second in-car study with a focus on a whole entertainment system (use case 2). Therefore, I will apply the social control concepts to the IVIS to regulate access to functions for both the driver and the passenger. This means I will investigate how access to IVIS functions can support driver-passenger collaboration. To evaluate this use case I will conduct a simulator study with driver-passenger pairs working together. With this experiment, I expect to understand how the social control concepts perform with regard to collaboration and social connectedness when applied to a whole entertainment system. Besides that, it will provide insights into how social control impacts safety-critical driving.

RQ1 & 2 – Social Control in the (Smart) Home: Currently, I am also working on a systematic literature review about the needs and motivations of (smart) home users. I conducted the scoping review with three databases (*ACM Digital Library*, *IEEE Explore*, *Scopus*) and identified 233 papers. After removing duplicates and

scanning the remained papers on title and abstract, 152 papers have been excluded resulting in a total of 81 papers. As a next step, I am reviewing these 81 papers with regard to motivational factors[38] and basic human needs [25]. Based on the detailed output, I will decide whether or not a follow-up study with regard to (smart) home users' needs and motivations is required. This overview and insights will further be used to concretize use case 1 (music playlist) and use case 2 (entertainment system) for the (smart) home domain. To apply and assess social control in the (smart) home domain, I will first systematically compare the use cases with the automotive use cases to be able to transfer the knowledge gained. To get feedback on the final social control setup I plan to organize expert sessions with experts from the remote control industry. Afterward, I will conduct an experiment per use case in a living room setting to assess the social control concepts with regard to collaboration and social connectedness for the (smart) home domain.

RQ3 – Generalization of Social Control for Entertainment Services: As a final step of my PhD program, my goal is to develop a social control framework that outlines technical design principles to support collaborative group decision-making and social connectedness. Therefore, I will compare the findings from the automotive studies with the results from the home studies under the lens of the strong concepts theory [17]. With this theory, I will extract technical design elements which go beyond particular domains (automotive, home) and thus have the potential to be appropriate for other services out of other domains and use cases.

7 EXPECTED CONTRIBUTION OF MY PHD PROGRAM

To summarize, my doctoral research contributes a framework to design for social control which outlines guidelines and design recommendations to distribute control in interactive systems for enhancing group work and social connectedness. Through experiments planned and conducted in two different domains (car and home), I evaluate(d) how users behave and perceive interactive systems that support social control and propose improvements based on these experiments. In a broader scope, the insights from my studies contribute to understanding people and interactive system design. Thus, researchers and designers from several research fields designing interactive systems for a social context will benefit from my work. Besides that, my insights also contribute to HCI research significantly, because more and more entertainment takes place in a social setting. Thus it is crucial to support social connectedness and to understand how control can be distributed across people in a social group setting. In addition, my studies contribute towards the automotive research field since I extended insights into passengers' needs and motivations and outline possibilities to support collaboration between people sitting in a car. With the focus on the living room environment, my findings also contribute to the design of future smart homes and entertainment technologies. Based on qualitative insights on the home experience, my studies will extend the design space for the next generation of (smart) home and entertainment services.

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